**From:**
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**Through:**
**Project Supervisor**
Department of Information Technology
E.G.S. Pillay Engineering College

**To:**
**The Head  Project Coordinator**
E.G.S. Pillay Engineering College
Nagapattinam – 611002

**Subject: Request for Additional Components and Materials Based on Updated Project Requirements**

Respected Sir/Madam,

I am **Ravikumar S**, a student of the Department of Information Technology, currently working on the final-year project titled **"IoT-Based Adaptive Sleep Monitoring and Environment Control System."**

During the recent external review of our project, the project implementation and requirements were further evaluated. Based on the suggestions and recommendations provided by the external reviewer, certain additional hardware components and supporting materials are required to update the project and improve its functionality, wearable implementation, automation, testing, and final prototype demonstration.

Therefore, I kindly request you to provide the required additional components and materials, or permit us to procure them, so that the suggested modifications can be implemented and the updated project requirements can be completed successfully.

The required components are attached/listed along with this request for your kind consideration and approval.

I kindly request your approval and necessary support for obtaining the above-mentioned requirements.

Thank you for your consideration and support.

**Yours faithfully,**

| No. | Category | Component | Model / Specification | Qty. | Purpose |
|---|---|---|---|---|---|
| 1 | Sensor | Microphone Module | MAX4466 | 1 | Environmental sound/noise monitoring |
| 2 | Sensor | PIR Motion Sensor | HC-SR501 | 1 | Room movement/presence detection |
| 3 | Controller | Microcontroller | ESP32 | 1 | Additional wearable/room node |
| 4 | Processing | Raspberry Pi | Raspberry Pi 4 – 4GB | 1 | Central processing and interface |
| 5 | Interface | Touch Display | 7" HDMI Touch Display | 1 | Local user interface |
| 6 | Interface | HDMI Cable | HDMI to HDMI | 1 | Raspberry Pi to display |
| 7 | Power | Li-Ion Battery | 3.7V 2500mAh | 1 | Wearable power supply |
| 8 | Power | Battery Charging Module | TP4056 | 1 | Battery charging |
| 9 | Power | Buck Converter | LM2596 | 1 | Voltage regulation |
| 10 | Actuator | Mini Exhaust Fan | 5V / 2A | 1 | Adaptive airflow control |
| 11 | Control | MOSFET | IRF540 | 1 | Fan/load switching |
| 12 | Alert | Buzzer | Standard | 1 | System alert |
| 13 | Storage | MicroSD Card | 32GB / 64GB Class 10 | 1 | Raspberry Pi OS and storage |
| 14 | Wiring | Jumper Wires | Male-to-Male (M-M) | 1 set | Circuit connections |
| 15 | Wiring | Jumper Wires | Male-to-Female (M-F) | 1 set | Sensor/module connections |
| 16 | Wiring | Jumper Wires | Female-to-Female (F-F) | 1 set | Module connections |

| No. | Category | Component | Model / Specification | Qty. | Purpose |
|---|---|---|---|---|---|
| 17 | Wiring | Pin Headers | Male + Female, 2.54mm | 1 set | Board/module connections |
| 18 | Components | Resistor Set | Assorted 1/4W | 1 set | Circuit connections |
| 19 | Components | Capacitor Set | Assorted | 1 set | Filtering/decoupling |
| 20 | Components | Diodes | 1N4007 | 5–10 | Circuit protection |
| 21 | Components | Terminal Connectors | 2-pin / 3-pin | 1 set | Power/actuator wiring |
| 22 | Power | 5V Power Supply | 5V, ≥2A | 1 | Fan/actuator power |
| 23 | Power | Power Switch | Mini ON/OFF | 1 | Wearable/system power |
| 24 | Mechanical | Watch Enclosure | Custom / 3D printed | 1 | Wearable electronics housing |
| 25 | Mechanical | Fan Guard / Mount | Suitable for 5V fan | 1 | Safe fan mounting |
| 26 | Mounting | Cable Ties | Small | 1 pack | Wire management |
| 27 | Mounting | Screws & Nuts | Assorted small sizes | 1 set | Mechanical assembly |